EXTERNAL AND INTERNAL MARKETS FOR MANAGERS

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MOTIVATION

- Finding, training and retaining managerial talent is a key determinant of firm productivity
  - Fundamental worker-input in production [Bender et al. (2018)]
  - Skill diffusion within firm [Minni (2023)]

- Two channels of manager (re)allocation
  - Between firm: Competition and poaching of managers
  - Within firm: Internal promotion + learning on the job
    - ~ 40% of inflows into management are from within the establishment

What are the implications of internal and external markets for managers in terms of:

- Individual outcomes? - Managerial Wage Premium, wage growth
- Aggregate managerial skill and its distribution across firms?

THIS PAPER

- Model
  - Search model with novel firm occupations and reallocation choices
  - External frictional flows depend crucially on how firms are currently organized
  - Worker Learning + Internal promotion $\rightarrow$ Avoid search frictions and keep high productivity

THIS PAPER

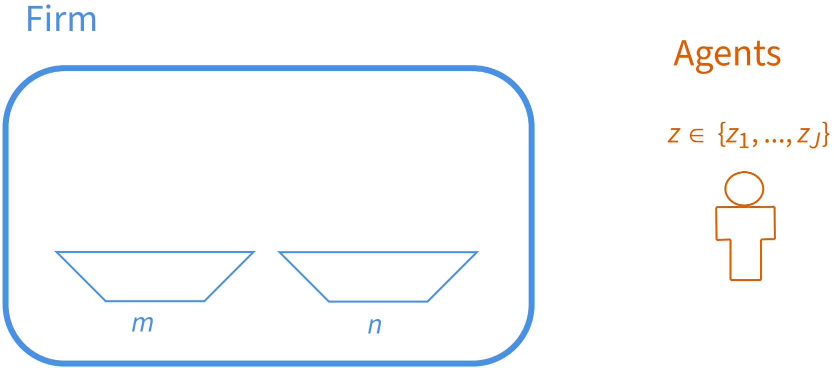
- Empirical Evidence
  - Managerial Wage Premium
  - Substantial internal inflow into management positions
  - Non-Managerial Learning with Managers
- Bringing model to data
  - Calibration of key parameters to target novel moments
    - Match the internal and external inflows into management
  - Policy: Targeted non-compete agreements for either managers or non-managers
    - Managers: Fast track promotions; middle-skilled agents become managers
    - Non-Managers: Delayed promotions; higher skill accumulation

CONTRIBUTION TO LITERATURE

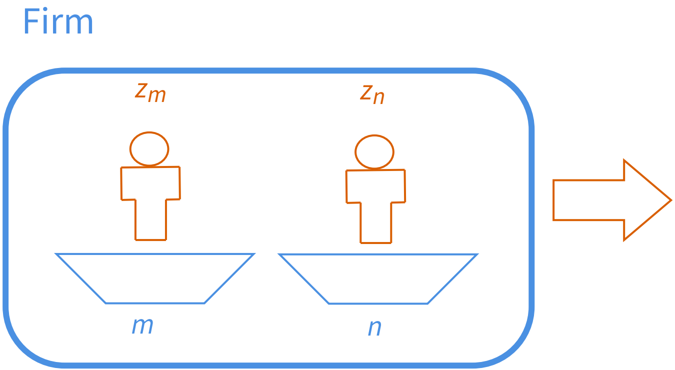
- Firm dynamics with frictional labor markets: Schaal (2017), Gouin-Bonenfant (2022), Bilal et al. (2022), Elsby and Gottfries (2022), Audoly (2023), Herkenhoff et al. (2024)
- Firm hierarchy and task division:Garicano and Rossi-Hansberg (2006), Caliendo et al. (2020), Adenbaum (2023), Kohlhepp (2023), Freund (2024)
- Managerial Alloc. and Productivity: Bloom and Van Reenen (2007), Minni (2023), Friedrich (2023), Pastorino (2022), Metcalfe, Sollaci, and Syverson (2023), Bender et al. (2018)

MODEL

PRIMITIVES

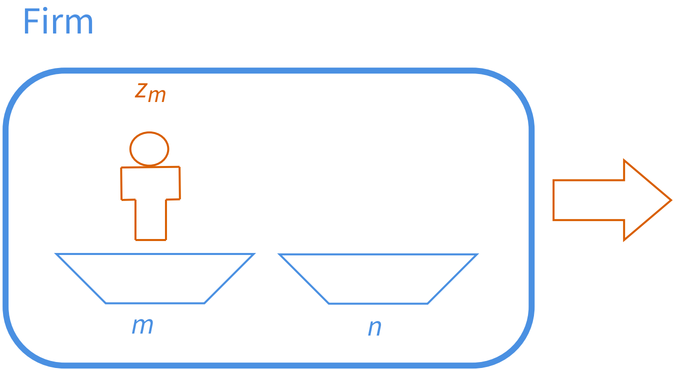


PRIMITIVES



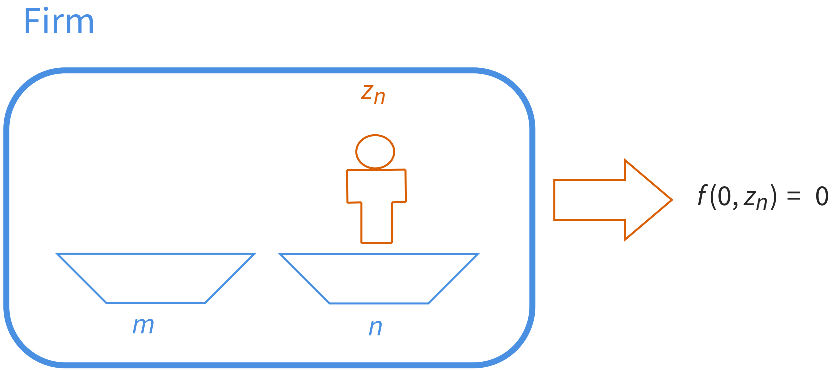
$$f(z_m, z_n) = z_m^{\alpha}\, z_n^{1-\alpha}$$

PRIMITIVES



$f(z_m, 0) = z_m^{\alpha}\, z_1^{1-\alpha}$

PRIMITIVES



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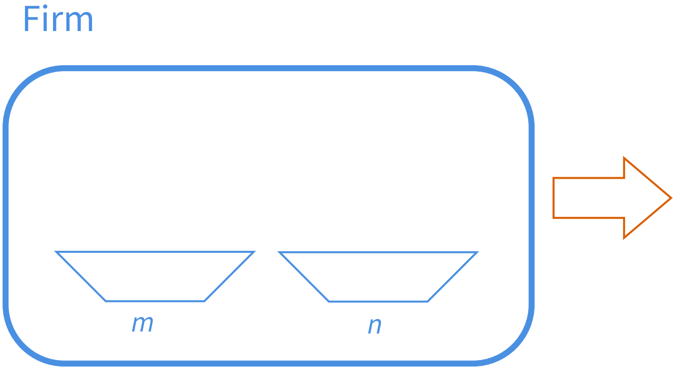

$f(0, 0) = 0$

TIMING

- Each period has four stages:
  1. Learning Shocks
  2. Search and Match ($\tilde{V}$)
  3. Reallocation ($\hat{V}$)
  4. Production and Wages (V)

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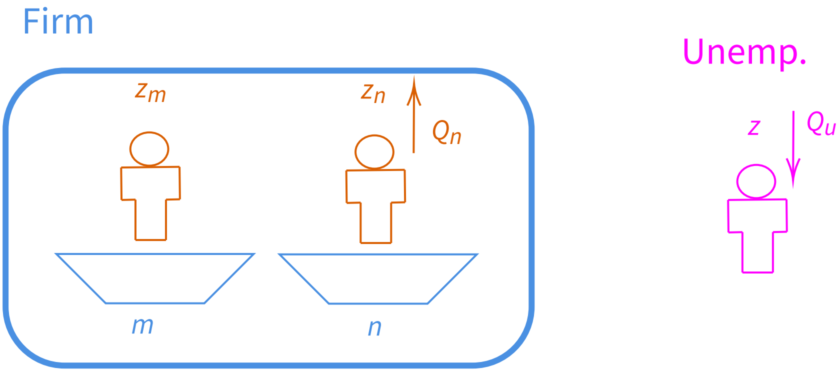
LEARNING SHOCK



$$Q_n\left(z_n' \middle| z_n, z_m\right) = \rho\frac{(z_m - z_n)^+}{z_J - z_1}$$

TIMING

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    1. Learning Shocks
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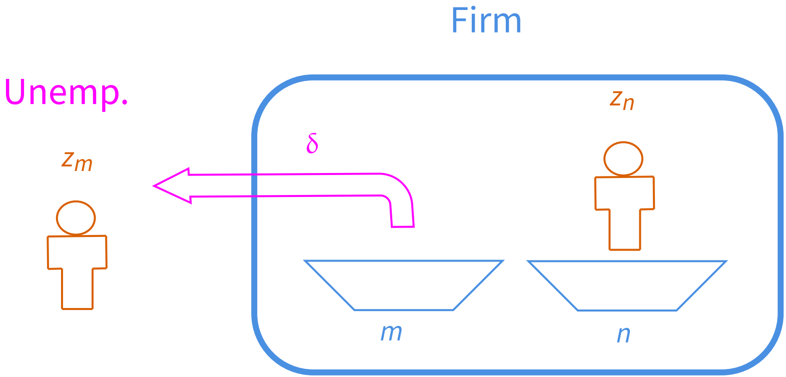
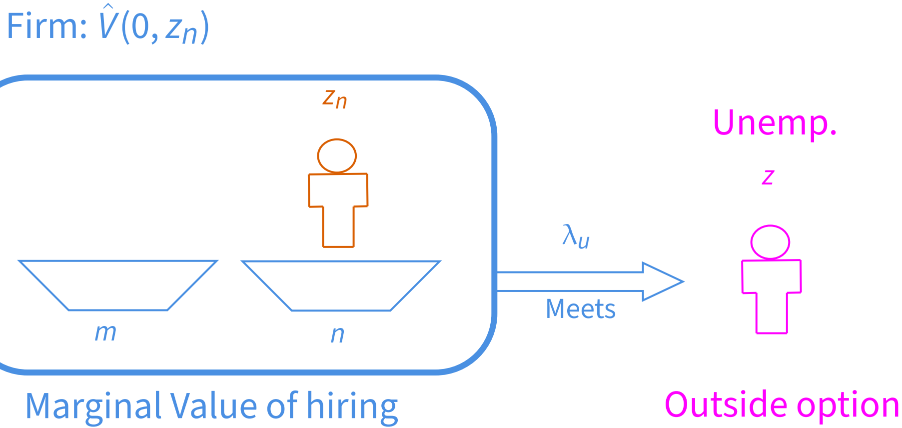
    4. Production and Wages (V)

SEARCH AND MATCH - SHOCK TO UNEMPLOYMENT

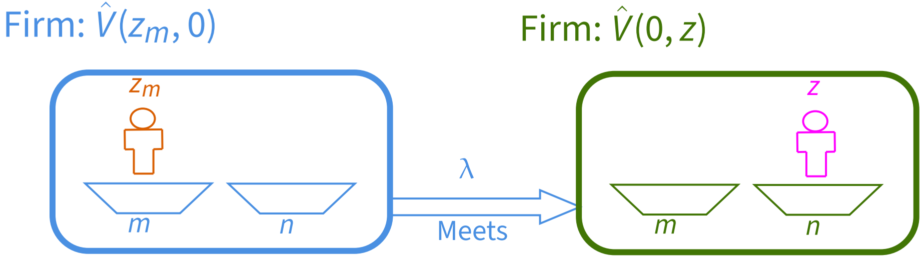


SEARCH AND MATCH - FIRM MEETS UNEMPLOYED



$$\max\left\{\hat{V}(0, z) + U(z_n), \hat{V}(z, z_n), \hat{V}(z_n, z) - c_p\right\} - \hat{V}(0, z_n) \quad - \quad U(z) \quad \geq 0$$

SEARCH AND MATCH - FIRM MEETS EMPLOYED AT ANOTHER FIRM



Marginal Value of hiring — **Outside option**

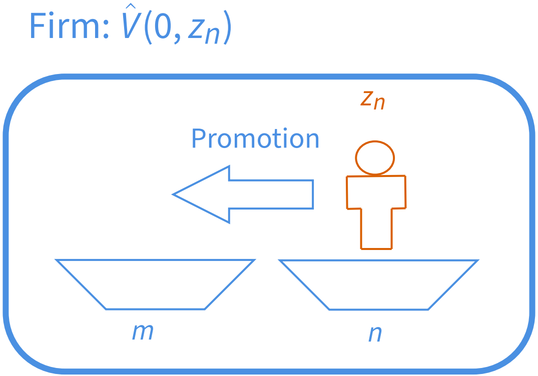
$$\max\left\{\hat{V}(z,0) + U(z_m), \hat{V}(z,z_m) - c_d\, , \hat{V}(z_m,z) \right\} - \hat{V}(z_m,0) \;-\; \left(\hat{V}(0,z) - \hat{V}(0,0)\right) \;\geq\; 0$$

◂ Marginal Values ◂ Outside Options

TIMING

- Each period has four stages:
  1. Learning Shocks
  2. Search and Match ($\tilde{V}$)
  3. Reallocation ($\hat{V}$)
  4. Production and Wages (V)

REALLOCATION



$$\max\left\{V(0, z_n),\ V(z_n, 0)\ - c_p\right\}$$

TIMING

- Each period has four stages:
    1. Learning Shocks
    2. Search and Match ($\tilde{V}$)
    3. Reallocation ($\hat{V}$)
    4. Production and Wages (V)

EMPLOYEE'S SHARE AND WAGES

- Upon hiring, employee gets share γ of the gains from trade
- Delivered via a wage, constant unless:
  1. Employee poached by another firm
  2. Incumbent keeps the employee after a poaching attempt
  3. Current employee value falls bellow the actual Marginal value
- In line with multi-worker search models Bilal et al. (2022) and Herkenhoff et al. (2024)
- Promotions are not necessarily immediately priced, only in cont. value
- Equilibrium is characterized independently of wages

DISTRIBUTIONS

- Let $e_u(z)$ be the mass of unemployed agents with productivity z
- Let $e(z_m, z_n)$ be the mass of firms with state (z_m, z_n)
- Distributions are key endogenous objects
  - Masses in each state depend on hiring and reallocation decisions
  - Distributions affect directly the continuation values of firms

VALUE FOR VACANT FIRMS

$$
\begin{aligned}
\tilde{V}(0,0) = \hat{V}(0,0) & \\
& + \sum_{z'} \frac{\lambda_u e_u(z')}{N}(1-\gamma)\left[\nu_{z'}(0,0) - u_{z'}\right]^{+} \quad \text{[Meet Unemp.]} \\
& + \sum_{y'_m} \frac{\lambda e(y'_m)}{N}(1-\gamma)\left[\nu_{z'}(0,0) - u_m(y'_m)\right]^{+} \quad \text{[Meet Firm w/ Manager]} \\
& + \sum_{y'_n} \frac{\lambda e(y'_n)}{N}(1-\gamma)\left[\nu_{q'}(0,0) - u_n(y'_n)\right]^{+} \quad \text{[Meet Firm w/ Worker]} \\
& + \sum_{y'_t} \frac{\lambda e(y'_t)}{N}(1-\gamma)\left[\max\left\{\nu_{z'}(0,0) - u_m(y'_t), \nu_{q'}(0,0) - u_n(y'_t)\right\}\right]^{+} \quad \text{[Meet Firm w/ a Team]}
\end{aligned}
$$

VALUE FOR FIRMS WITH A TEAM

$$\begin{aligned}
\tilde{V}(z_m, z_n) = {} & \hat{V}(z_m, z_n) \\
& + \delta\left[\hat{V}(0, z_n) + U(z_m) - \hat{V}(z_m, z_n)\right] + \delta\left[\hat{V}(z_m, 0) + U(z_n) - \hat{V}(z_m, z_n)\right] \\
& + \sum_{z'} \frac{\lambda_u e_u(z')}{N}(1-\gamma)\left[\nu_{z'}(z_m, z_n) - u_{z'}\right]^+ \text{ [Meet Unemp.]} \\
& + \sum_{y'_m} \frac{\lambda e(y'_m)}{N}(1-\gamma)\left[\nu_{z'}(z_m, z_n) - u_m(y'_m)\right]^+ \text{ [Meet Firm w/ Manager]} \\
& + \sum_{y'_n} \frac{\lambda e(y'_n)}{N}(1-\gamma)\left[\nu_{z'}(z_m, z_n) - u_n(y'_n)\right]^+ \text{ [Meet Firm w/ Worker]} \\
& + \sum_{y'_t} \frac{\lambda e(y'_t)}{N}(1-\gamma)\left[\max\left\{\nu_{z'}(z_m, z_n) - u_m(y'_t), \nu_{q'}(z_m, z_n) - u_n(y'_t)\right\}\right]^+ \text{ [Meet Firm w/ a Tear} \\
& + \sum_{y} \frac{\lambda e(y)}{N}\gamma\left[\max\left\{\nu_{z_m}(y) - u_m(z_m, z_n), \nu_{z_n}(y) - u_n(z_m, z_n)\right\}\right]^+ \text{ [Poached by another firm]}
\end{aligned}$$

◂ Unempl. Workers ◂ Firms w/ Manager ◂ Firms w/ Worker

REALLOCATION

- After the SM stage, firms can reallocate current workers
- Given production values $V(y)$ and unemployment value $U(z_m)$, the firm solves

$$\hat{V}(0,0) = V(0,0)$$

$$\hat{V}(z_m,0) = \max\left\{V(z_m,0), V(0,0) + U(z_m), V(0,z_m) - c_d\right\}$$

$$\hat{V}(0,z_n) = \max\left\{V(0,z_n), V(0,0) + U(z_n), V(z_n,0) - c_p\right\}$$

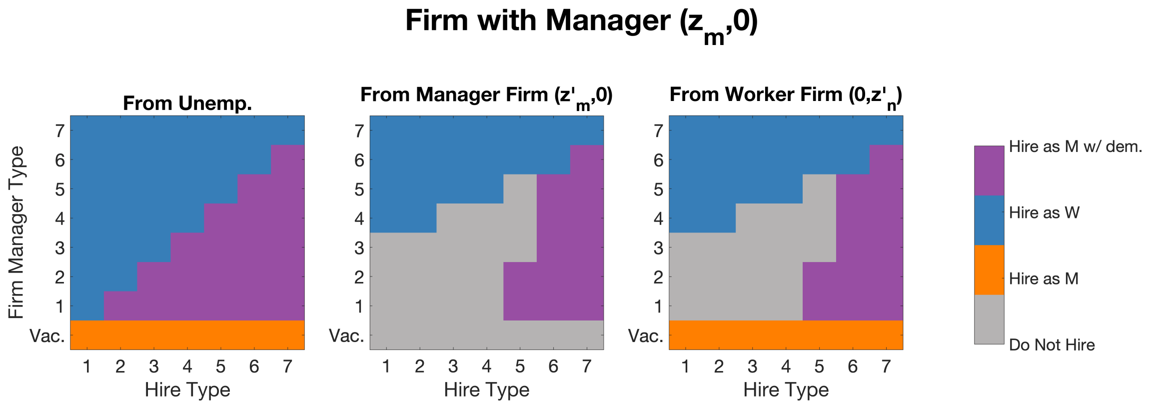
$$\begin{aligned}\hat{V}(z_m,z_n) = \max\Big\{&V(z_m,z_n), V(z_m,0) + U(z_n), V(0,z_n) + U(z_m), V(z_n,z_m) - c_d - c_p,\\ &V(0,z_m) + U(z_n) - c_d, V(z_n,0) + U(z_m) - c_p, V(0,0) + U(z_m) + U(z_n)\Big\}\end{aligned}$$

◂ Production

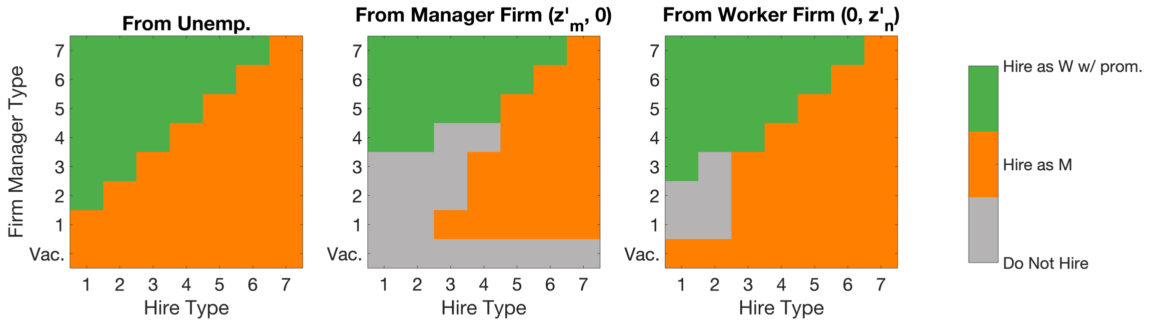
EQUILIBRIUM

- A Stationary Equilibrium is:
  - A set of values $\left\{U(z), \hat{V}(y), V(y)\right\}$ that satisfy the Bellman equations, given the distributions $\left\{e_u(z), e(y)\right\}$
  - The distribution $\left\{e_u(z), e(y)\right\}$ is stationary given the policies implied by the values functions

- Reminder: Wages inverted from equilibrium values and policies


Firm with Manager (z_m,0)
From Unemp.
From Manager Firm (z'_m,0)
From Worker Firm (0,z'_n)
Firm Manager Type
7
6
5
4
3
2
1
Vac.
1 2 3 4 5 6 7
Hire Type
Hire as M w/ dem.
Hire as W
Hire as M
Do Not Hire

Firm with Worker $(0,z_n)$



EMPIRICAL EVIDENCE

LIAB - WORKER-ESTABLISHMENT DATA

- Entire biographies of the complete workforce of a sample of establishments
- Organized in a yearly panel 2013-2017 (Dauth and Eppelsheimer (2020)) ◂ Cleaning
- 5-digit occupation codes (*KLdB 2010*)
  - Overall occupation
  - Hierarchy: Manager and Non-Manager
  - Task complexity

◂ Year Dist. ◂ Some Numbers on Panel Cases ◂ Examples

EMPIRICAL FACTS

1. Wages ◂ Specifics
   - Managers are paid ~ 14% more than comparable non-managers
     (complexity, occupation, firm size, age, tenure, education, industry, region)
2. Flows ◂ Decompostion
   - 40% of inflows into managers are internal
   - Same magnitude of external inflows into managerial positions
3. Non-Managerial Learning from Managers

◂ Add. Empirics

NON-MANAGERIAL LEARNING

- Restrict to non-managers with a job-loss episode, transitioned to another establishment
  (Jarosch, Oberfield, and Rossi-Hansberg (2021), Herkenhoff et al. (2024))
- How much does the average wage of my previous manager affects my next wage?
- Managers in the same 3-digit occupation as *i* at *t*

$$\text{lwage}_{i,t+1} = \beta_0 + \beta\,\text{lwage}_{i,t} + \theta\ \text{M Wage}_{i,t} + \text{Controls} + \epsilon_{i,t}$$

- Controls & FE: Task complexity, occupation, tenure, education, industry, region, year

β	θ
0.330	0.071
(0.00)	(0.03)

◂ Add. Spec.

CALIBRATION

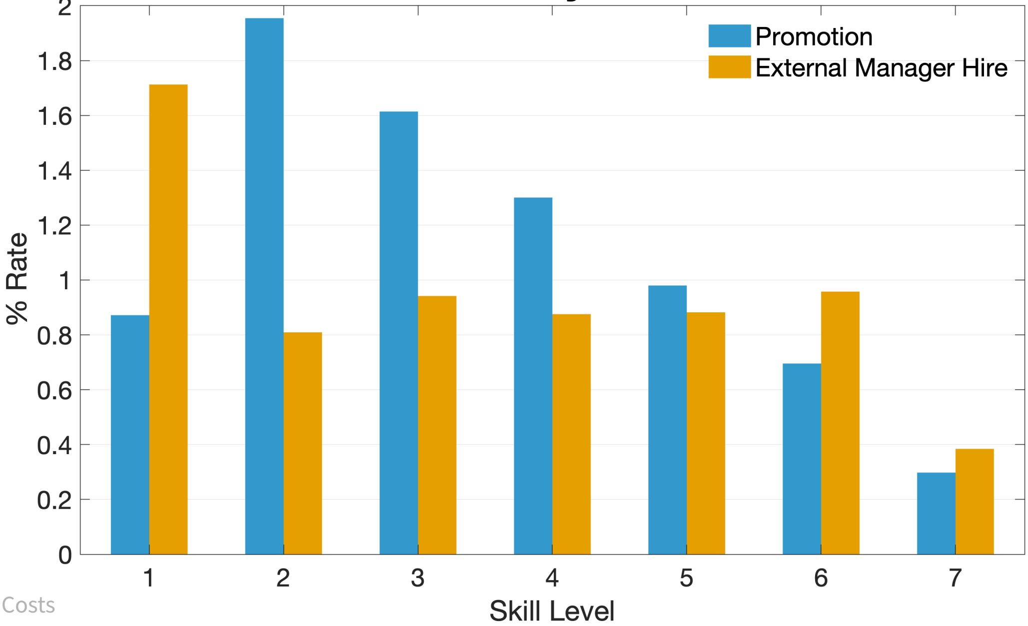
Parameters	Description	Targets	Values
c_p	Promotion Cost (% avg prod.)	Promotion rate	18.39%
c_d	Demotion Cost (% avg prod.)	Demotion rate	18.68%
α	Manager Share	M. wage premium	0.683
Q_{up}	W Learning w/ M	θ_1, θ_2 Learning reg.	0.047
λ	Meeting rate	External M hire rate	0.034
λ_u	Unemp. Meeting rate	Unemp. to M hire rate	0.058
δ	Separation rate	M to Unemp. rate	0.001

◂ Pre-set Parameters

MODEL FIT

Moments	**Data**	**Model**
Promotion rate	0.47%	0.51%
Demotion rate	0.11%	0.25%
External M hire rate	0.41%	0.56%
Unemp. to M hire rate	0.20%	0.87%
M to Unemp. rate	0.07%	0.73%
M. wage premium	0.14	0.15
θ_1 Learning	0.33	0.37
θ_2 Learning	0.07	0.07

Flow Rates by Skill Level



◂ Diff Costs

NON-COMPETE AGREEMENTS (NCAS)

- Use the model as a lab to study the effects of non-compete agreements
- Unique to our setting:
  - Targeted NCAs to managers and non-managers
  - Role of internal reallocation in the effects of NCAs
- NCAs in the model:
  - Upon meeting a firm, prob. φ_m (managers) and φ_n (non-managers) that the firm has a NCA on that position
  - Adjust probabilities to get average duration of NCAs
  - Today: 12 months duration; individually targeted

% Baseline	NCA Manager	NCA Worker
Manager Skill	1.37	0.21
Manager Wage	-0.87	-1.05
Worker Skill	-4.77	1.21
Worker Wage	-0.94	-3.37
Average Product	-1.42	0.70

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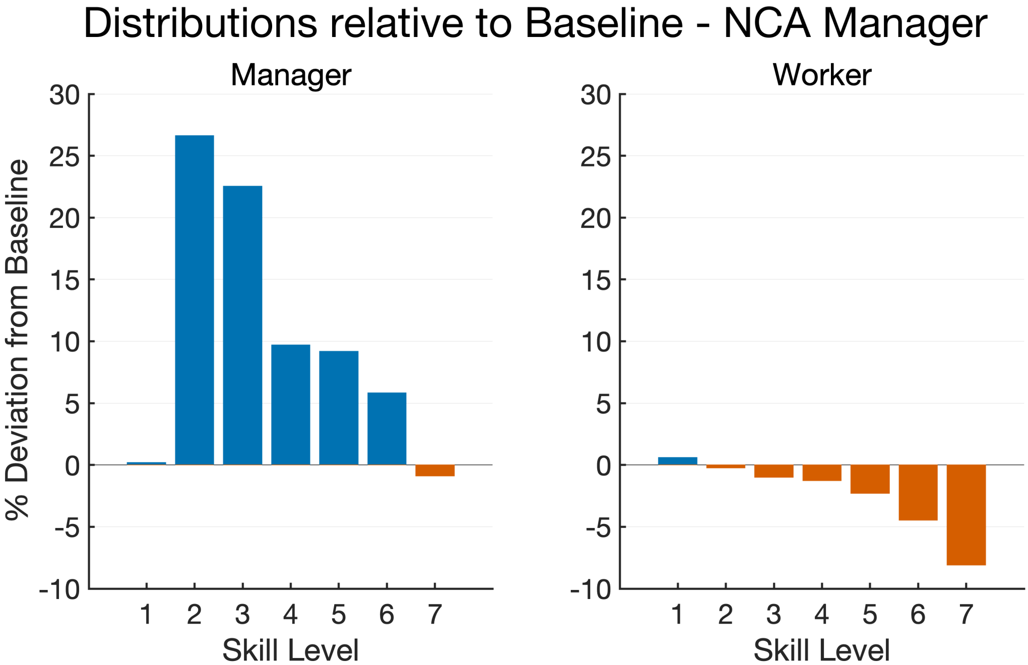
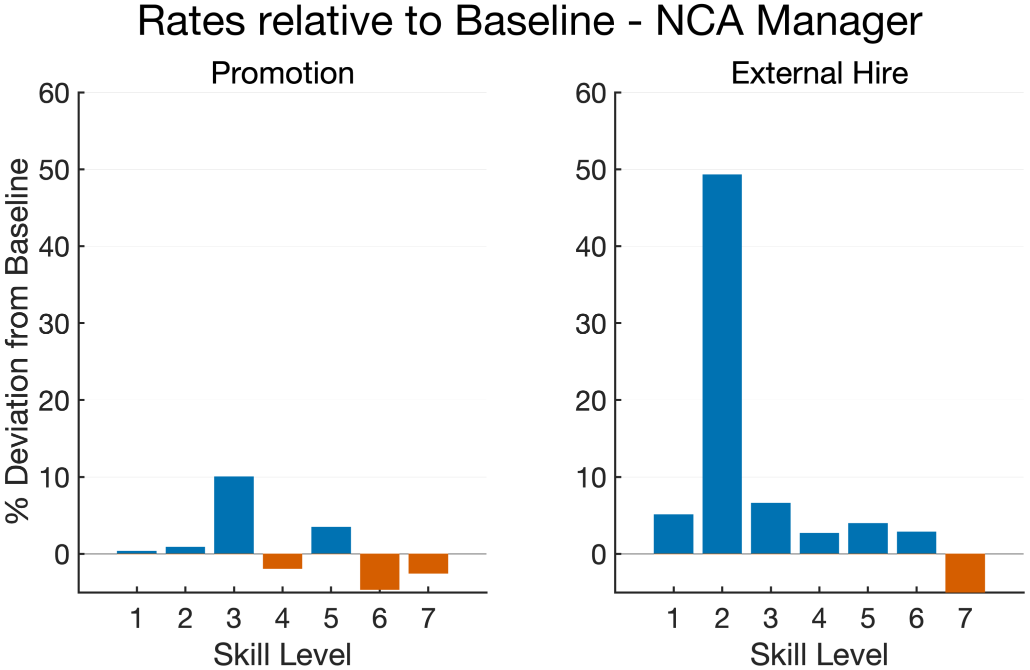

Distributions relative to Baseline - NCA Manager
Manager
Worker
% Deviation from Baseline
30
25
20
15
10
5
0
-5
-10
1
2
3
4
5
6
7
Skill Level
Skill Level



Rates relative to Baseline - NCA Manager
Promotion
External Hire
% Deviation from Baseline
60
50
40
30
20
10
0
1
2
3
4
5
6
7
Skill Level
Skill Level

% Baseline	NCA Manager	NCA Worker
Manager Skill	1.37	0.21
Manager Wage	-0.87	-1.05
Worker Skill	-4.77	1.21
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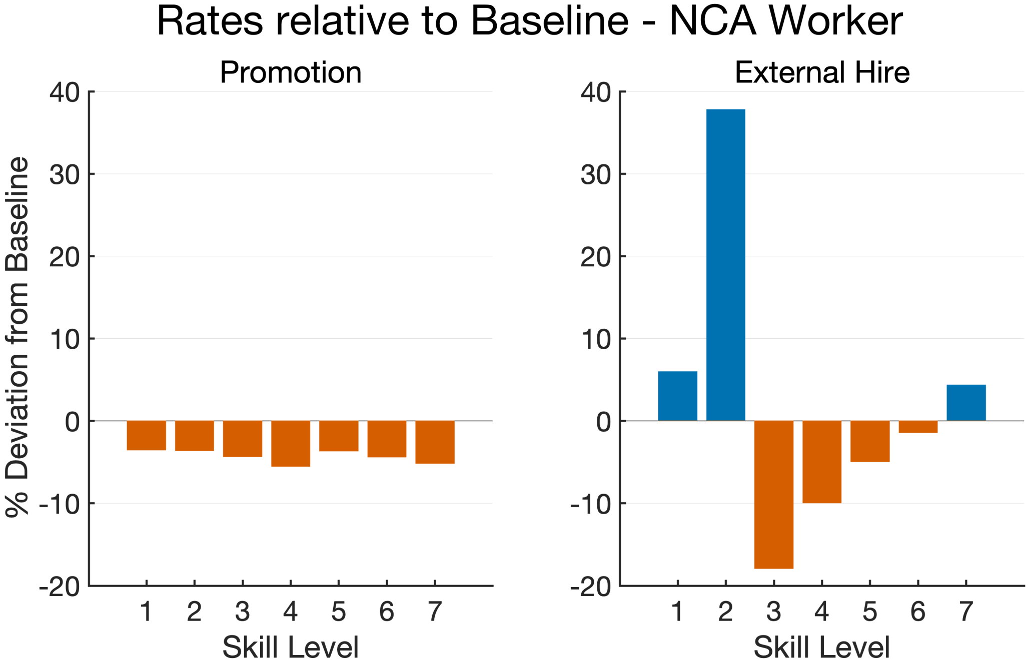
Average Product	-1.42	0.70



Rates relative to Baseline - NCA Worker
Promotion
External Hire
% Deviation from Baseline
40
30
20
10
0
-10
-20
1
2
3
4
5
6
7
Skill Level

% Baseline	NCA Manager	NCA Worker
Manager Skill	1.37	0.21
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NON-COMPETE AGREEMENTS

- NCA on managers:
  - Fast track promotions, specially on lower skill levels
  - Shift skill away from workers, making their wages also falls
  - Product falls, with less skill accumulation and more low skill managers
- NCA on non-managers:
  - Promotion rates fall for all skill levels
  - Workers accumulate skill, but still see wage compression
  - Product can increase, as overall skill in the economy rises

IN TODAY'S TALK

- Search model with internal reallocation and learning:
  - Mechanism to keep productivity high in face of manager turnover
- In the data Managers are:
  - Highly paid workers; associated with worker learning
  - Substantially promoted from within the establishment
- Next steps:
  - Additional decomposition and exercises on the NCAs counterfactuals
  - Wage growth around internal and external hires ◂ in the data
  - Research Agenda: Explore mechanisms of firm structure in shaping internal allocation under external competition.

Thank You!

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CONTRIBUTION TO LITERATURE

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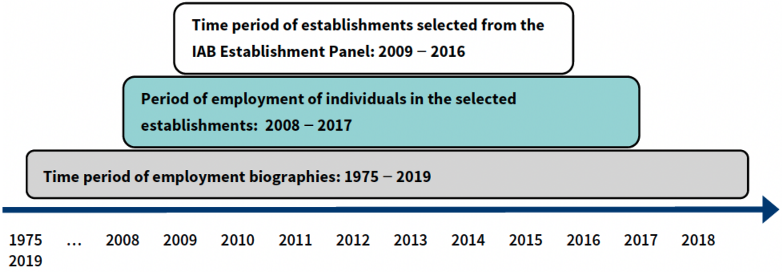
DATA APPENDIX

DATA CLEANING

- Cleaning process following Dauth and Eppelsheimer (2020)
- Merge with BHP (firm data)
- Generate Industries 2 digit, Occ and educational consolidation groups
- Wages: Deflated to 2015 Euros
- Flagging top-censored wages
- Have available an imputed wage following Card, Heining, and Kline (2013), Dustmann, Ludsteck, and Schönberg (2009)
- Yearly panel with main episode and overlapping Jan 31st (spell-level data also available)
- Construct *panel cases* from establishments that were surveyed by the, where we have the entire workforce of the firm

DATA SOURCES

- LIAB LM 1975-2019



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SOME NUMBERS ON THE PANEL CASES

- Overall
  - From 2009–2016, ~ 7k firms, ~ 600k workers per year
  - 2-digit Industry: Manufacturing (23%), Trade & Repair (15%), Real Estate (12%), Construction (8%)
- Managers
  - ~ 35k managers per year, around 5% of the workforce
  - 2-digit Industry: Manufacturing (25%), Trade & Repair (17%), Real Estate (11%), Construction (9%)
  - Manager + High Complexity: ~ 20k per year (after 2011)
  - Similar pattern of industries, but more heavily concentrated in healthcare and education

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EXAMPLE KLDB 2010 AND ISCO-08 COMPARISON

KldB 2010 (5-Digit)	KldB 2010 Classification title	ISCO-08 (4-Digit)	ISCO Unit Group
27394	Managers in production planning and scheduling	1321	Manufacturing managers
28194	Managers in textile making	1321	Manufacturing managers
28294	Managers in the production of clothing and other textile products	1321	Manufacturing managers
28394	Managers in leather- and fur-making and -processing	1321	Manufacturing managers
29194	Managers in beverages production	1321	Manufacturing managers
29294	Managers in the production of foodstuffs (etc)	1321	Manufacturing managers
82594	Managers in medicine, orthopaedic and rehabilitation technology	1321	Manufacturing managers

EXAMPLE KLDB 2010 AND ISCO-08 COMPARISON

KldB 2010 (5-Digit)	Classification title	ISCO-08 (4-Digit)	ISCO-08 (4-Digit)
29301	Cooks (without specialization)-unskilled/semiskilled tasks	9412	Kitchen helpers
29302	Cooks (without specialization)-skilled tasks	5120	Cooks
29312	Hors d'œuvrier, pantry or pastry cooks-skilled tasks	5120	Cooks
29322	Roast, grill or fish cooks-skilled tasks	5120	Cooks
29382	Cooks (with specialization, not elsewhere classified)-skilled tasks	5120	Cooks
29393	Supervisors in cooking	3434	Chefs
29394	Managers in cooking	3434	Chefs

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MANAGERS ON THE PANEL CASES

Category	2009	2010	2011	2012	2013	2014	2015	2016	Total
Non-Managers	593,817	582,362	571,662	573,434	579,256	589,614	610,520	626,717	4,727,382
Managers	22,101	22,666	31,192	37,441	37,745	37,694	37,906	38,533	265,278
(%)	3.72%	3.89%	5.46%	6.53%	6.52%	6.39%	6.21%	6.15%	5.61%

FIRMS WITH AT LEAST ONE MANAGER

Category	2009	2010	2011	2012	2013	2014	2015	2016	Total
Firms w/o Managers	4,221	4,562	4,150	4,117	4,360	4,673	5,077	5,671	36,831
Firm with Managers	1,945	1,941	2,445	2,646	2,712	2,780	2,864	2,976	20,309
Percentage (%)	46.08%	42.55%	58.92%	64.27%	62.20%	59.49%	56.41%	52.48%	55.14%

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INFLOWS INTO MANAGERS AND NON-MANAGERS

From	Into *M*	Into *NM*
M same firm	7.50%	1.02%
M diff firm	15.80%	1.02%
NM same firm	36.13%	12.70%
NM diff firm	21.86%	41.50%
Unemp.	18.50%	43.60%

INFLOWS INTO MANAGERS AND NON-MANAGERS (TOTAL WORKERS)

From	Into *M*	Into *NM*
M same firm	0.90%	0.17%
M diff firm	2.10%	0.17%
NM same firm	4.70%	2.10%
NM diff firm	2.80%	7.10%
Unemp.	2.40%	7.50%

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OUTFLOWS OUT OF MANAGERS AND NON-MANAGERS

<table>
<tr><th>To</th><th>Out of M</th><th>Out of NM</th><th>Out of Unemp</th></tr>
<tr><td>M same firm</td><td>0.99%</td><td>0.29%</td><td rowspan="2">2.43%</td></tr>
<tr><td>M diff firm</td><td>2.09%</td><td>0.18%</td></tr>
<tr><td>NM same firm</td><td>2.80%</td><td>2.19%</td><td rowspan="2">7.53%</td></tr>
<tr><td>NM diff firm</td><td>2.82%</td><td>7.16%</td></tr>
<tr><td>Unemp.</td><td>0.79%</td><td>1.22%</td><td></td></tr>
</table>

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MANAGERS IN THE CROSS-SECTION

- Simplest regression

$$\text{lwage}_{it} = \beta_0 + \beta\text{Manager}_{it} + \sum_{1}^{4} \beta_n \text{Complex}_{it} + \Gamma X_{it} + \epsilon_{it}$$

- Controls: Age, Tenure, Education, Occupation, Industry, Region, Firm Size

	(1)	(2)	(3)	(4)
Manager	0.672	0.609	0.141	0.149
	(0.00)	(0.00)	(0.00)	(0.00)
Skill level 2			0.184	0.176
			(0.00)	(0.00)
Skill level 3			0.444	0.390
			(0.00)	(0.00)
Skill level 4			0.587	0.500
			(0.00)	(0.00)
Year FE	Yes	Yes	Yes	Yes
Ind. Demographics		Yes	Yes	Yes
Firm Chact.			Yes	Yes

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NON-MANAGER LEARNING

- Look at NM to M transitions to another firm, *through an unemployment spell*
- Key control: previous firm's Avg. Wage (measure of how productive it was)

$$\text{lwage}_{i,t+2} = \beta_0 + \beta \text{lwage}_{i,t} + \gamma \bar{W}_{i,t} + \text{controls} + \epsilon_{i,t}$$

	NM-M diff firm	NM-U-M	NM-U-M (FE)
β	0.42	0.233	0.168
	(0.00)	(0.00)	(0.00)
γ	0.116	0.198	0.143
	(0.006)	(0.05)	(0.223)

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MODEL APPENDIX

MARGINAL VALUES

- Firm in state y marginal value of hiring an agent z'

$$
v_{z'}(y) = \begin{cases}
\max\left\{\hat{V}(z', 0), \hat{V}(0, z')\right\} - \hat{V}(0,0) & \text{if } y = (0,0) \\
\max\left\{\hat{V}(z', 0) + U(z_m), \hat{V}(z_m, z'), \hat{V}(z', z_m) - c_d\right\} - \hat{V}(z_m, 0) & \text{if } y = (z_m, 0) \\
\max\left\{\hat{V}(z', z_n), \hat{V}(z_n, z') - c_p, \hat{V}(0, z') + U(z_n)\right\} - \hat{V}(0, z_n) & \text{if } y = (0, z_n) \\
\max\left\{\hat{V}(z', z_n) + U(z_m), \hat{V}(z', z_m) + U(z_n) - c_d, \hat{V}(z_m, z') + U(z_n), \right. & \\
\qquad\qquad \left. \hat{V}(z_n, z') + U(z_m) - c_p\right\} - \hat{V}(z_m, z_n) & \text{if } y = (z_m, z_n)
\end{cases}
$$

OUTSIDE OPTION

- Outside option of a *manager* z' in state y' is

$$u_{m,z'}(y') = \begin{cases} \hat{V}(z',0) - \hat{V}(0,0) & \text{if } y' = (z',0) \\ \hat{V}(z',z_n) - \hat{V}(0,z_n') & \text{if } y' = (z',z_n') \end{cases}$$

- Outside option of a *worker* z' in state y' is

$$u_{n,z'}(y') = \begin{cases} \hat{V}(0,z') - \hat{V}(0,0) & \text{if } y' = (0,z') \\ \hat{V}(z_m',z') - \hat{V}(z_m',0) & \text{if } y' = (z_m',z') \end{cases}$$

- Outside option of an *unemployed worker* z' is simply: $u_{z'} = U(z')$

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VALUE OF UNEMPLOYED WORKERS

$$U(z_m) = b(z_m) + \beta \mathbb{E}_{z'_m} \left[U(z'_m) + \sigma[0 - U(z'_m)] + (1 - \sigma) \left[\sum_y \frac{\lambda_u e(y)}{N} \gamma \left[\nu_{z'_m}(y) - U(z'_m) \right]^+ \right] \right]$$

- At the production stage, receives the home production $b(z_m)$
- Suffer shocks of productivity and leave the model with probability σ
- If not, meet a firm and receive a fraction γ of the gains from trade

VALUE FOR FIRMS WITH MANAGER

$$
\begin{aligned}
\tilde{V}(z_m, 0) = \hat{V}(z_m, 0) + \delta \left[\hat{V}(0,0) + U(z_m) - \hat{V}(z_m, 0)\right] \\
&+ \sum_{\tilde{z}_m} \frac{\lambda_u e_u(\tilde{z}_m)}{N}(1-\gamma)\left[\nu_{\tilde{z}_m}(z_m, 0) - u_{\tilde{z}_m}\right]^+ \\
&+ \sum_{(\tilde{a},\tilde{z}_m)} \frac{\lambda e(\tilde{a},\tilde{z}_m,0)}{N}(1-\gamma)\left[\nu_{\tilde{z}_m}(z_m, 0) - u_{m,\tilde{z}_m}(\tilde{a}, \tilde{z}_m, 0)\right]^+ \\
&+ \sum_{(\tilde{a},\tilde{z}_n)} \frac{\lambda e(\tilde{a},0,\tilde{z}_n)}{N}(1-\gamma)\left[\nu_{\tilde{z}_n}(z_m, 0) - u_{n,\tilde{z}_n}(\tilde{a}, 0, \tilde{z}_n)\right]^+ \\
&+ \sum_{(\tilde{a},\tilde{z}_m,\tilde{z}_n)} \frac{\lambda e(\tilde{a},\tilde{z}_m,\tilde{z}_n)}{N}(1-\gamma)\left[\max\left\{\nu_{\tilde{z}_m}(z_m, 0) - u_{m,\tilde{z}_m}(\tilde{a}, \tilde{z}_m, \tilde{z}_n), \nu_{\tilde{z}_n}(z_m, 0) - u_{n,\tilde{z}_n}(\tilde{a}, \tilde{z}_m, \right.\right. \\
&+ \sum_{y} \frac{\lambda e(y)}{N}\gamma\left[\nu_z(y) - u_{m,z_m}(z_m, 0)\right]^+
\end{aligned}
$$

VALUE FOR FIRMS WITH WORKER

$$
\begin{aligned}
\tilde{V}(0, z_n) = \hat{V}(0, z_n) + \delta \left[\hat{V}(0,0) + U(z_n) - \hat{V}(0, z_n)\right] \\
&+ \sum_{\tilde{z}_m} \frac{\lambda_u e_u(\tilde{z}_m)}{N}(1-\gamma)\left[\nu_{\tilde{z}_m}(0, z_n) - u_{\tilde{z}_m}\right]^+ \\
&+ \sum_{(\tilde{a},\tilde{z}_m)} \frac{\lambda e(\tilde{a},\tilde{z}_m,0)}{N}(1-\gamma)\left[\nu_{\tilde{z}_m}(0, z_n) - u_{m,\tilde{z}_m}(\tilde{a},\tilde{z}_m,0)\right]^+ \\
&+ \sum_{(\tilde{a},\tilde{z}_n)} \frac{\lambda e(\tilde{a},0,\tilde{z}_n)}{N}(1-\gamma)\left[\nu_{\tilde{z}_n}(0, z_n) - u_{n,\tilde{z}_n}(\tilde{a},0,\tilde{z}_n)\right]^+ \\
&+ \sum_{(\tilde{a},\tilde{z}_m,\tilde{z}_n)} \frac{\lambda e(\tilde{a},\tilde{z}_m,\tilde{z}_n)}{N}(1-\gamma)\left[\max\left\{\nu_{\tilde{z}_m}(0, z_n) - u_{m,\tilde{z}_m}(\tilde{a},\tilde{z}_m,\tilde{z}_n), \nu_{\tilde{z}_n}(0, z_n) - u_{n,\tilde{z}_n}(\tilde{a},\tilde{z}_m,\tilde{z}\right.\right. \\
&+ \sum_{y} \frac{\lambda e(y)}{N}\gamma\left[\nu_{z_n}(y) - u_{n,z_n}(0, z_n)\right]^+
\end{aligned}
$$

PRODUCTION

- Value functions at the production stage are

$$V(0,0) = f(0,0) + \beta \mathbb{E}_{a'} \left[\tilde{V}(0,0) \right]$$

$$V(z_m,0) = f(z_m,0) + \beta \mathbb{E}_{a'} \left[\tilde{V}(z_m,0) \right]$$

$$V(0,z_n) = f(0,z_n) + \beta \mathbb{E}_{q'} \left[\tilde{V}(0,z_n') \right]$$

$$V(z_m,z_n) = f(z_m,z_n) + \beta \mathbb{E}_{q'} \left[\tilde{V}(z_m,z_n') \right]$$

◂ Back

PRODUCTION

- Value functions at the production stage are

$$
\begin{aligned}
V(0,0) &= f(0,0) + \beta\left[\sum_{a'} \pi(a'|a)\tilde{V}(0,0)\right] \\
V(z_m,0) &= f(z_m,0) + \beta\left[\sum_{a'} \pi(a'|a)\left(\sigma\tilde{V}(0,0) + (1-\sigma)\tilde{V}(z_m,0)\right)\right] \\
V(0,z_n) &= f(0,z_n) + \beta\left[\sum_{z_n'} \pi(a'|a)Q(z_n'|z_n,a)\left(\sigma\tilde{V}(0,0) + (1-\sigma)\tilde{V}(0,z_n')\right)\right] \\
V(z_m,z_n) &= f(z_m,z_n) + \beta\left[\sum_{z_n'} \pi(a'|a)Q(z_n'|z_n,a)\left(\sigma^2\tilde{V}(0,0) + \right.\right. \\
&\qquad \left.\left.\sigma(1-\sigma)\tilde{V}(z_m,0) + \sigma(1-\sigma)\tilde{V}(0,z_n') + (1-\sigma)^2\tilde{V}(z_m,z_n')\right)\right]
\end{aligned}
$$

POLICY FUNCTIONS - HIRINGS

- The gains from trade, pin down the hiring function at the SM stage
- Firm y hires z' from
  - From Unemployment

$$h_{uz'}(y) = \mathbb{1}\left\{\nu_{z'}(y) - u_{z'} > 0\right\}$$

  - From from with Manager $(z', 0)$

$$h_{m(z')}(y) = \mathbb{1}\left\{\nu_{z'}(y) - u_{m,z'}(z', 0) > 0\right\}$$

  - From from with Worker $(0, z')$

$$h_{n(z')}(y) = \mathbb{1}\left\{\nu_{z'}(y) - u_{n,z'}(0, z') > 0\right\}$$

POLICY FUNCTIONS - HIRINGS

- The gains from trade, pin down the hiring function at the SM stage
- Firm y hires z' from
  - From Firm with Team (z', q'), hire the manager

$$h_{m(z',q')}(y) = \mathbb{1}\left\{\nu_{z'}(y) - u_{m,z'}(z',q') > \left[\nu_{q'}(y) - u_{n,q'}(z',q')\right]^{+}\right\}$$

  - From Firm with Team (z', q'), hire the worker

$$h_{n(z',q')}(y) = \mathbb{1}\left\{\nu_{q'}(y) - u_{n,q'}(z',q') > \left[\nu_{z'}(y) - u_{m,z'}(z',q')\right]^{+}\right\}$$

POLICY FUNCTIONS - ALLOCATIONS

- Conditional on hiring z', the firm decides where to allocate the worker based on continuation values $\hat{V}$
- Empty firm $(0, 0)$ allocates the hire z' as
  - As Manager
$$p_{m,z'}(0,0) = \mathbb{1}\left\{\hat{V}(z',0) \geq \hat{V}(0,z')\right\}$$
  - As Worker $p_{n,z'}(0,0) = 1 - p_{m,z'}(0,0)$

◂ Back

POLICY FUNCTIONS - ALLOCATIONS

- Conditional on hiring z', the firm decides where to allocate the worker based on continuation values $\hat{V}$
- Firm with Manager $(z_m, 0)$ allocates the hire z' as
  - As Manager, firing the current manager

$$p^u_{m,z'}(z_m, 0) = \mathbb{1}\left\{\hat{V}(z', 0) + U(z_m) > \max\left\{\hat{V}(z_m, z'), \hat{V}(z', z_m) - c_d\right\}\right\}$$

  - As manager, demoting the current manager

$$p^d_{m,z'}(z_m, 0) = \mathbb{1}\left\{\hat{V}(z', z_m) - c_d > \max\left\{\hat{V}(z_m, z'), \hat{V}(z', 0) + U(z_m)\right\}\right\}$$

  - As Worker $p_{n,z'}(z_m, 0) = 1 - p_{m,z'} - p^d_{m,z'}$

◂ Back

POLICY FUNCTIONS - ALLOCATIONS

- Firm with Worker $(0, z_n)$ allocates the hire z' as
  - As Worker, firing the current worker

$$p^u_{n,z'}(0, z_n) = \mathbb{1}\left\{\hat{V}(0, z') + U(z_n) > \max\left\{\hat{V}(z', z_n), \hat{V}(z_n, z') - c_p\right\}\right\}$$

  - As Worker, promoting the current worker

$$p^p_{n,z'}(0, z_n) = \mathbb{1}\left\{\hat{V}(z_n, z') - c_p > \max\left\{\hat{V}(z', z_n), \hat{V}(0, z') + U(z_n)\right\}\right\}$$

  - As Manager $p^u_{m,z'}(0, z_n) = 1 - p_{n,z'} - p^p_{n,z'}$

POLICY FUNCTIONS - ALLOCATIONS

- Firm with Team (z_m, z_n) allocates the hire z' as
  - As Manager, firing the current manager

$$p^u_{m,z'}(z_m, z_n) = \mathbb{1}\left\{\hat{V}(z', z_n) + U(z_m) > \max\left\{\hat{V}(z_m, z') + U(z_n), \hat{V}(z', z_m) + U(z_n) - c_d, \hat{V}(z_n, z') + U(z_m) - c_p\right\}\right\}$$

  - As Manager, demoting the current manager

$$p^d_{m,z'}(z_m, z_n) = \mathbb{1}\left\{\hat{V}(z', z_m) + U(z_n) - c_d > \max\left\{\hat{V}(z', z_n) + U(z_m), \hat{V}(z_m, z') + U(z_n), \hat{V}(z_n, z') + U(z_m) - c_p\right\}\right\}$$

POLICY FUNCTIONS - ALLOCATIONS

- Firm with Team (z_m, z_n) allocates the hire z' as
  - As Worker, firing the current worker

$$p^u_{n,z'}(z_m, z_n) = \mathbb{1}\left\{\hat{V}(z_m, z') + U(z_n) > \max\left\{\hat{V}(z', z_n) + U(z_m), \hat{V}(z', z_m) + U(z_n) - c_d, \hat{V}(z_n, z') + U(z_m) - c_p\right\}\right\}$$

  - As Worker, promoting the current worker

$$p^p_{n,z'}(z_m, z_n) = \mathbb{1}\left\{\hat{V}(z_n, z') + U(z_m) - c_p > \max\left\{\hat{V}(z', z_n) + U(z_m), \hat{V}(z', z_m) + U(z_n) - c_d, \hat{V}(z_m, z') + U(z_n)\right\}\right\}$$

◂ Back

POLICY FUNCTIONS - REALLOCATION

- Post SM and before production, any firm can decide to reallocate or fire its workers
- Firm with manager $(z_m, 0)$ can
  - Fire the manager

$$d_m(z_m, 0) = \mathbb{1}\left\{V(0,0) + U(z_m) > \max\left\{V(0, z_m) - c_d, V(z_m, 0)\right\}\right\}$$

  - Demote the manager

$$r_m(z_m, 0) = \mathbb{1}\left\{V(0, z_m) - c_d > \max\left\{V(0,0) + U(z_m), V(z_m, 0)\right\}\right\}$$

  - Or stay Idle with $1 - d_m(z_m, 0) - r_m(z_m, 0)$

◂ Back

POLICY FUNCTIONS - REALLOCATION

- Firm with worker $(0, z_n)$ can
  - Fire the worker

$$d_n(0, z_n) = \mathbb{1}\left\{V(0,0) + U(z_n) > \max\left\{V(0, z_n) - c_p, V(z_n, 0)\right\}\right\}$$

  - Promote the worker

$$r_n(0, z_n) = \mathbb{1}\left\{V(z_n, 0) - c_p > \max\left\{V(0,0) + U(z_n), V(0, z_n)\right\}\right\}$$

  - Or stay Idle with $1 - d_n(0, z_n) - r_n(0, z_n)$

◂ Back

POLICY FUNCTIONS - REALLOCATION

- Firm with team (z_m, z_n) have to decide what to do with both agents
  - Fire manager only

$$
\begin{aligned}
d_m(z_m, z_n) = \mathbb{1}\Big\{\max\big\{&V(0, z_n) + U(z_m), V(z_n, 0) + U(z_m) - c_p\big\} \\
> \max\big\{&V(z_m, z_n), V(z_m, 0) + U(z_n), V(0, z_m) + U(z_n) - c_d, \\
&V(z_n, z_m) - c_d - c_p, V(0, 0) + U(z_m) + U(z_n)\big\}\Big\}
\end{aligned}
$$

POLICY FUNCTIONS - REALLOCATION

- Firm with team (z_m, z_n) have to decide what to do with both agents
  - Fire worker only

$$\begin{aligned} d_n(z_m, z_n) = \mathbb{1}\Big\{ &\max\left\{V(z_m, 0) + U(z_n), V(0, z_m) + U(z_n) - c_d\right\} \\ &> \max\left\{V(z_m, z_n), V(z_n, 0) + U(z_m), V(0, z_n) + U(z_m) - c_p,\right. \\ &\qquad \left. V(z_n, z_m) - c_d - c_p, V(0, 0) + U(z_m) + U(z_n)\right\}\Big\} \end{aligned}$$

◂ Back

POLICY FUNCTIONS - REALLOCATION

- Firm with team (z_m, z_n) have to decide what to do with both agents
  - Reallocate manager

$$r_m(z_m, z_n) = \mathbb{1}\left\{\max\left\{V(0, z_m) + U(z_n) - c_d, V(z_n, z_m) - c_p - c_d\right\} > \max\{\ldots\}\right\}$$

  - Reallocate worker

$$r_n(z_m, z_n) = \mathbb{1}\left\{\max\left\{V(z_n, 0) + U(z_m) - c_p, V(z_n, z_m) - c_p - c_d\right\} > \max\{\ldots\}\right\}$$

POLICY FUNCTIONS - REALLOCATION

- Firm with team (z_m, z_n) have to decide what to do with both agents
  - Firm both agents

$$d_{mn}(z_m, z_n) = \mathbb{1}\left\{\max\left\{V(0,0) + U(z_m) + U(z_n)\right\} > \max\{\dots\}\right\}$$

  - Stay idle with remaining probability
  - We have a complete description of events with the above policy functions

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LAW OF MOTIONS

◂ Back

WAGES

- Given the equilibrium value and policy functions, we can write the agent's value functions
- The share of the worker in the surplus is delivered by a wage that remains constant, *unless* to induce a correct allocation across employment states
- As in Herkenhoff et al. (2024) wage only changes when:
    1. The agent is poached by another firm, gets its Mg Value at the new firm
    2. The incumbent firm keeps the agent after a poaching attempt, raises the wage to match the agent's outside option
    3. The current value of the agent falls below the Mg Value of the agent to the firm at its current position

◂ Back

WAGES

- The first two conditions raise wages to ensure the correct allocation of the agent across firms
- The third condition lowers the wage, adjusting for the worker's actual marginal value to the firm. This ensures that agents always have incentives to report offers truthfully to the team.
- Assumptions are in line with the search framework and to some extent with the data
- Lack of competitive pressure on wages makes them sticky, and firms find optimal to adjust only when necessary

◂ Back

WAGES

- Important caveat: promotions are not immediately priced in the wage, but are in the continuation value of the agent
- When approached by another firm, the incumbent will value the promoted agetn as its Mg Value is higher
- How does that square off with data? Not sure and we are ill-suited to answer that question with the data we have
- On a broad sense it might be consistent with the apparent fact that internal promotions have a smaller premium than external hires into manager
- The alternative would be some ad-hoc assumption on how the wage change upon promotion, that would lack some contractual foundation (I think so)
- Since this paper is not about promotion premium we might be better off being a bit more agnostic about it, and sticking to what makes sense in a search framework

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AGENTS S&M VALUE FUNCTIONS

- For a manager in a firm $(z_m, 0)$
- Let

$$\mathbb{W}_m(z', w, z_m, 0) = p^u_{m,z'} U(z_m) + p^d_{m,z'} \hat{W}_n(w, z', z_m) \\ + (1 - p^u_{m,z'} - p^d_{m,z'}) \hat{W}_m(w, z_m, z') - \hat{W}_m(w, z_m, 0)$$

- Continuation for the manager z_m upon hiring z', considering all possible allocations
- Agent value function is

AGENTS S&M VALUE FUNCTIONS

- For a manager in a firm $(z_m, 0)$

$$
\begin{aligned}
\tilde{W}_m(w, z_m, 0) = \hat{W}_m(w, z_m, 0) &+ \delta \left[U(z_m) - \hat{W}_m(w, z_m, 0) \right] \\
&+ \sum_{z'} \frac{\lambda_u e_u(z')}{N} h_{uz'}(z_m, 0) \mathbb{W}_m(z', w, z_m, 0) \\
&+ \sum_{(z')} \frac{\lambda e(z',0)}{N} h_{m(z',0)}(z_m, 0) \mathbb{W}_m(z', w, z_m, 0) \\
&+ \sum_{(q')} \frac{\lambda e(0,q')}{N} h_{n(q')}(z_m, 0) \mathbb{W}_m(q', w, z_m, 0) \\
&+ \sum_{(z',q')} \frac{\lambda e(z',q')}{N} \left[h_{m(z',q')} \mathbb{W}_m(z', w, z_m, 0) + h_{n(z',q')} \mathbb{W}_m(q', w, z_m, 0) \right] \\
&+ \sum_{y} \frac{\lambda e(y)}{N} \left[\max \left\{ \hat{W}_m(w, z_m, 0), \min \left\{ \nu_{z_m}(y), \gamma \nu_{z_m}(y) + (1-\gamma) u_m(z_m, 0) \right\} \right\} \right. \\
&\qquad \left. - \hat{W}_m(w, z_m, 0) \right]
\end{aligned}
$$

AGENTS S&M VALUE FUNCTIONS

- For a worker in a firm $(0, z_n)$

$$
\begin{aligned}
\tilde{W}_n(w,0,z_n) = \hat{W}_n(w,0,z_n) &+ \delta\left[U(z_n) - \hat{W}_n(w,0,z_n)\right] \\
&+ \sum_{z'} \frac{\lambda_u e_u(z')}{N} h_{uz'}(0,z_n)\mathbb{W}_n(z',w,0,z_n) \\
&+ \sum_{(z')} \frac{\lambda e(z',0)}{N} h_{m(z',0)}(0,z_n)\mathbb{W}_n(z',w,0,z_n) \\
&+ \sum_{(q')} \frac{\lambda e(0,q')}{N} h_{n(q')}(0,z_n)\mathbb{W}_n(q',w,0,z_n) \\
&+ \sum_{(z',q')} \frac{\lambda e(z',q')}{N} \left[h_{m(z',q')}\mathbb{W}_n(z',w,0,z_n) + h_{n(z',q')}\mathbb{W}_n(q',w,0,z_n)\right] \\
&+ \sum_{y} \frac{\lambda e(y)}{N} \left[\max\left\{\hat{W}_n(w,0,z_n), \min\left\{\nu_{z_n}(y), \gamma\nu_{z_n}(y) + (1-\gamma)u_n(0,z_n)\right\}\right\}\right. \\
&\qquad \left. - \hat{W}_n(w,0,z_n)\right]
\end{aligned}
$$

AGENTS S&M VALUE FUNCTIONS

- For a manager in a firm (z_m, z_n)

$$
\begin{aligned}
\tilde{W}_m(w, z_m, z_n) = {} & \hat{W}_m(w, z_m, z_n) + \delta \left[U(z_m) - \hat{W}_m(w, z_m, z_n) \right] \\
& + \left(\delta + \sum_y \frac{\lambda e(y)}{N} h_{n(z_m, z_n)}(y) \left[\hat{W}_m(w, z_m, 0) - \hat{W}_m(w, z_m, z_n) \right] \right) \\
& + \sum_{z'} \frac{\lambda_u e_u(z')}{N} h_{uz'}(z_m, z_n) \mathbb{W}_m(z', w, z_m, z_n) \\
& + \sum_{(z')} \frac{\lambda e(z', 0)}{N} h_{m(z', 0)} \mathbb{W}_m(z', w, z_m, z_n) + \sum_{(q')} \frac{\lambda e(0, q')}{N} h_{n(q')} \mathbb{W}_m(q', w, z_m, z_n) \\
& + \sum_{(z', q')} \frac{\lambda e(z', q')}{N} \left[h_{m(z', q')} \mathbb{W}_m(z', w, z_m, z_n) + h_{n(z', q')} \mathbb{W}_m(q', w, z_m, z_n) \right] \\
& + \sum_y \frac{\lambda e(y)}{N} \Big[\max \left\{ \hat{W}_m(w, z_m, z_n), \min \left\{ \nu_{z_m}(y), \gamma \nu_{z_m}(y) + (1 - \gamma) u_m(z_m, z_n) \right\} \right\} \\
& \qquad - \hat{W}_m(w, z_m, z_n) \Big]
\end{aligned}
$$

AGENTS S&M VALUE FUNCTIONS

- For a worker in a firm (z_m, z_n)

$$
\begin{aligned}
\tilde{W}_n(w, z_m, z_n) = \hat{W}_n(w, z_m, z_n) &+ \delta \left[U(z_n) - \hat{W}_n(w, z_m, z_n) \right] \\
&+ \left(\delta + \sum_y \frac{\lambda e(y)}{N} h_{m(z_m, z_n)}(y) \left[\hat{W}_n(w, 0, z_n) - \hat{W}_n(w, z_m, z_n) \right] \right) \\
&+ \sum_{z'} \frac{\lambda_u e_u(z')}{N} h_{uz'}(z_m, z_n) \mathbb{W}_n(z', w, z_m, z_n) \\
&+ \sum_{(z')} \frac{\lambda e(z', 0)}{N} h_{m(z', 0)} \mathbb{W}_n(z', w, z_m, z_n) + \sum_{(q')} \frac{\lambda e(0, q')}{N} h_{n(q')} \mathbb{W}_n(q', w, z_m, z_n) \\
&+ \sum_{(z', q')} \frac{\lambda e(z', q')}{N} \left[h_{m(z', q')} \mathbb{W}_n(z', w, z_m, z_n) + h_{n(z', q')} \mathbb{W}_n(q', w, z_m, z_n) \right] \\
&+ \sum_y \frac{\lambda e(y)}{N} \left[\max \left\{ \hat{W}_n(w, z_m, z_n), \min \left\{ \nu_{z_n}(y), \gamma \nu_{z_n}(y) + (1 - \gamma) u_n(z_m, z_n) \right\} \right\} \right. \\
&\qquad \left. - \hat{W}_n(w, z_m, z_n) \right]
\end{aligned}
$$

POST SM AGENT'S VALUE FUNCTIONS

- For a manager in a firm $(z_m, 0)$

$$\begin{aligned}\hat{W}_m(w, z_m, 0) = \max \Big\{ & U(z_m),\;\; r_m \hat{W}_n(w, 0, z_m) \\ & + (1 - r_m) \min \Big\{ d_m U(z_m) + (1 - d_m) \hat{W}_m(w, z_m, 0), V(z_m, 0) - V(0, 0) \Big\} \Big\}\end{aligned}$$

- For a worker in a firm $(0, z_n)$

$$\begin{aligned}\hat{W}_n(w, 0, z_n) = \max \Big\{ & U(z_n),\;\; r_n \hat{W}_m(w, z_n, 0) \\ & + (1 - r_n) \min \Big\{ d_n U(z_n) + (1 - d_n) \hat{W}_n(w, 0, z_n), V(0, z_n) - V(0, 0) \Big\} \Big\}\end{aligned}$$

POST SM AGENT'S VALUE FUNCTIONS

- For a manager in a firm (z_m, z_n)

$$
\begin{aligned}
\hat{W}_m(w, z_m, z_n) = \max \Big\{ & U(z_m), d_n \left(r_m W_n(w, 0, z_m) + (1 - r_m) W_m(w, z_m, z_n) \right) \\
& + r_m r_n W_n(w, z_n, z_m) \\
& + (1 - d_n - r_n r_m) \min \Big\{ (d_m + d_{mn}) U(z_m) + (1 - d_m - d_{mn}) W_m(w, z_m, z_n), \\
& \qquad V(z_m, z_n) - V(0, z_n) \Big\} \Big\}
\end{aligned}
$$

POST SM AGENT'S VALUE FUNCTIONS

- For a worker in a firm (z_m, z_n)

$$\begin{aligned}\hat{W}_n(w, z_m, z_n) = \max \Big\{ & U(z_n), d_m \left(r_n W_m(w, z_n, 0) + (1 - r_n) W_n(w, 0, z_n) \right) + \\ & + r_n r_m W_m(w, z_n, z_m) \\ & + (1 - d_m - r_n r_m) \min \Big\{ (d_n + d_{mn}) U(z_n) + (1 - d_n - d_{mn}) W_n(w, z_m, z_n), \\ & \qquad V(z_m, z_n) - V(z_m, 0) \Big\} \Big\}\end{aligned}$$

PRODUCTIONS AND SHOCKS VALUE FUNCTIONS

- For manager in a firm $(z_m, 0)$

$$W_m(w, z_m, 0) = w + \beta \left[\sum_{a'} \pi(a'|a)(1-\sigma)\tilde{W}_m(w, a', z_m, 0)\right]$$

- For worker in a firm $(0, z_n)$

$$W_n(w, 0, z_n) = w + \beta \left[\sum_{a'}\sum_{z_n'} \pi(a'|a)Q(z_n'|z_n)(1-\sigma)\tilde{W}_n(w, a', 0, z_n')\right]$$

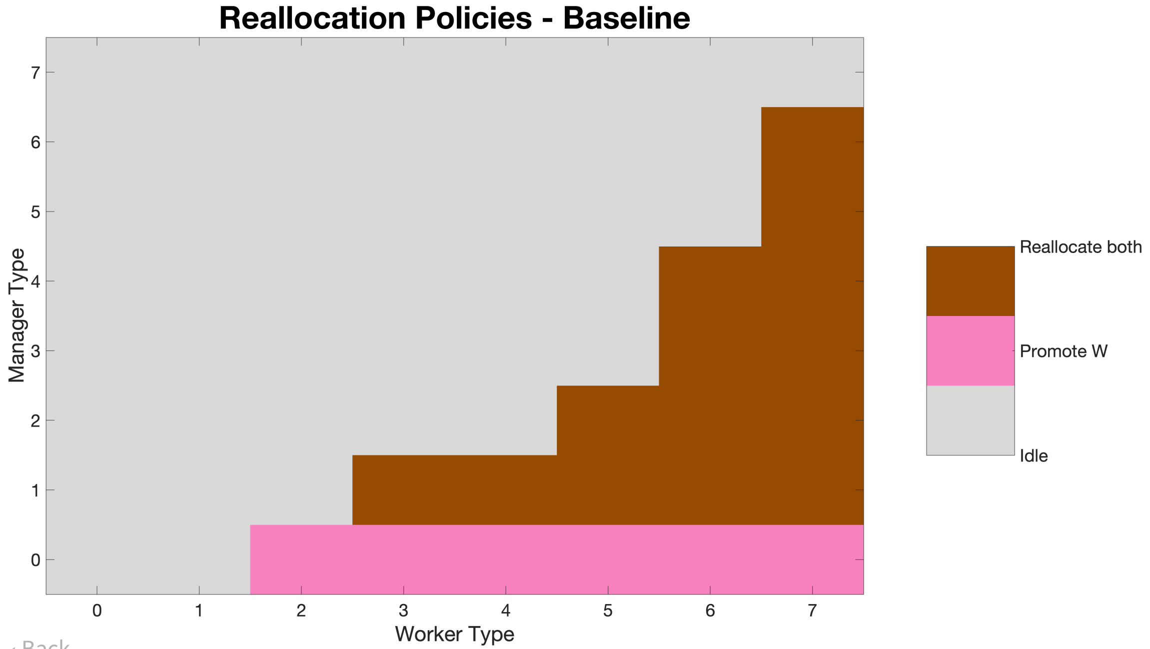
PRODUCTIONS AND SHOCKS VALUE FUNCTIONS

- For manager in a firm (z_m, z_n)

$$W_m(w, z_m, z_n) = w + \beta \left[\sum_{a'} \sum_{z_n'} \pi(a'|a) Q(z_n'|z_n) \left(\sigma(1-\sigma)\tilde{W}_m(w, a', z_m, 0) + (1-\sigma)^2 \tilde{W}_m(w, a', z_m, z_n') \right) \right]$$

- For worker in a firm (z_m, z_n)

$$W_n(w, z_m, z_n) = w + \beta \left[\sum_{a'} \sum_{z_n'} \pi(a'|a) Q(z_n'|z_n) \left(\sigma(1-\sigma)\tilde{W}_n(w, a', 0, z_n) + (1-\sigma)^2 \tilde{W}_n(w, a', z_m, z_n') \right) \right]$$


Reallocation Policies - Baseline
Manager Type
Worker Type
0
1
2
3
4
5
6
7
Reallocate both
Promote W
Idle
◂ Back

NON-MANAGERIAL LEARNING

- Restrict to non-managers with a job-loss episode, transitioned to another establishment (Jarosch, Oberfield, and Rossi-Hansberg (2021), Herkenhoff et al. (2024))
- How much average wage of my previous manager affects my next wage?

$$\text{lwage}_{i,t+1} = \beta_0 + \beta\,\text{lwage}_{i,t} + \theta \bar{W}_{i,t} + \text{Controls} + \epsilon_{i,t}$$

- Controls & FE: Task complexity, occupation, tenure, education, industry, region, year

	Avg M Wage	Same Occup.	Top M Wage Same Occ.
β	0.361	0.330	0.329
	(0.00)	(0.00)	(0.00)
θ	0.080	0.071	0.320
	(0.02)	(0.03)	(0.01)

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PRE-SET PARAMETERS

Parameters	Description	Values	Source
χ	Dispersion of newborns	2.6234	HLMP
$Q_{\overline{u}}$	Unemp. Skill Depreciation	0.016	HLMP
b	Flow value of unemployment	0.71	Freund (2024)
β	Discount Factor	0.992	10% annual rate
σ	Exit probability	0.0021	40 years in the mkt.
γ	Employee share of gains	0.5	Standard

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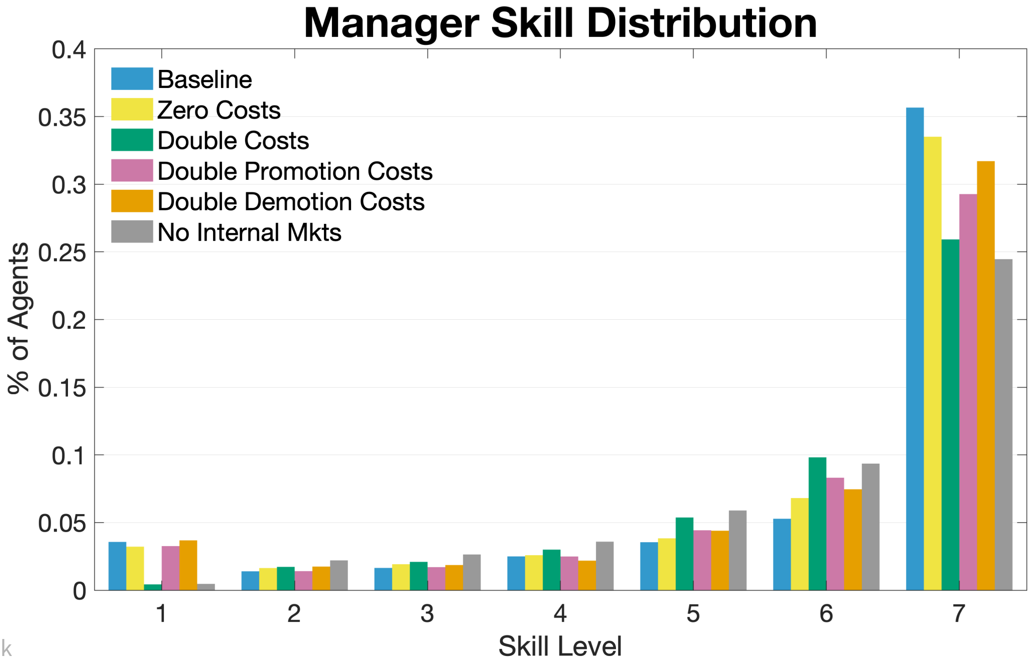
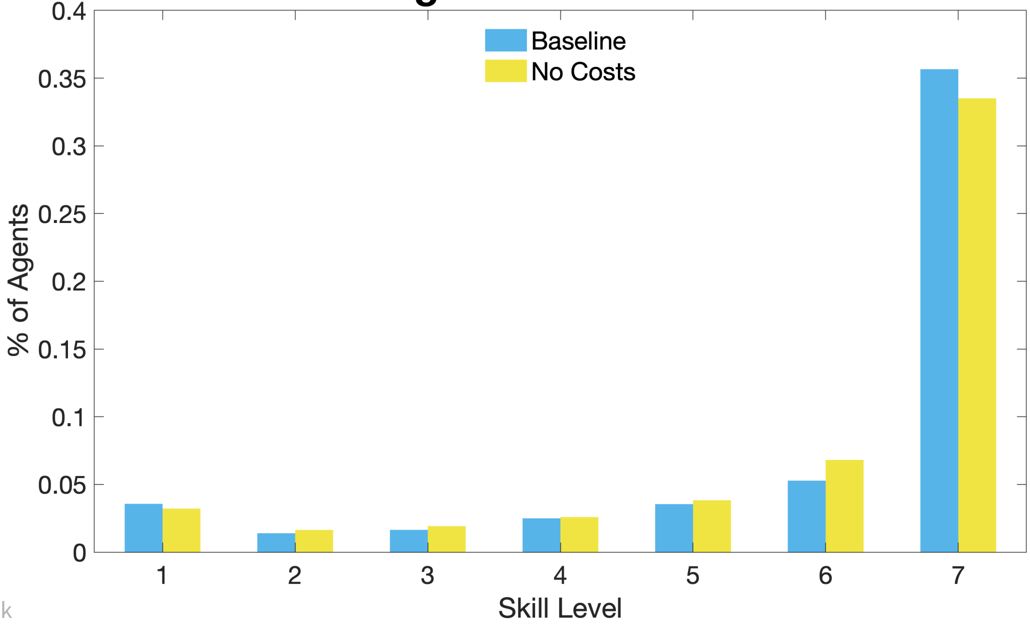

Manager Skill Distribution
Baseline
Zero Costs
Double Costs
Double Promotion Costs
Double Demotion Costs
No Internal Mkts
% of Agents
Skill Level
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
1
2
3
4
5
6
7

◂ Back

Manager Skill Distribution



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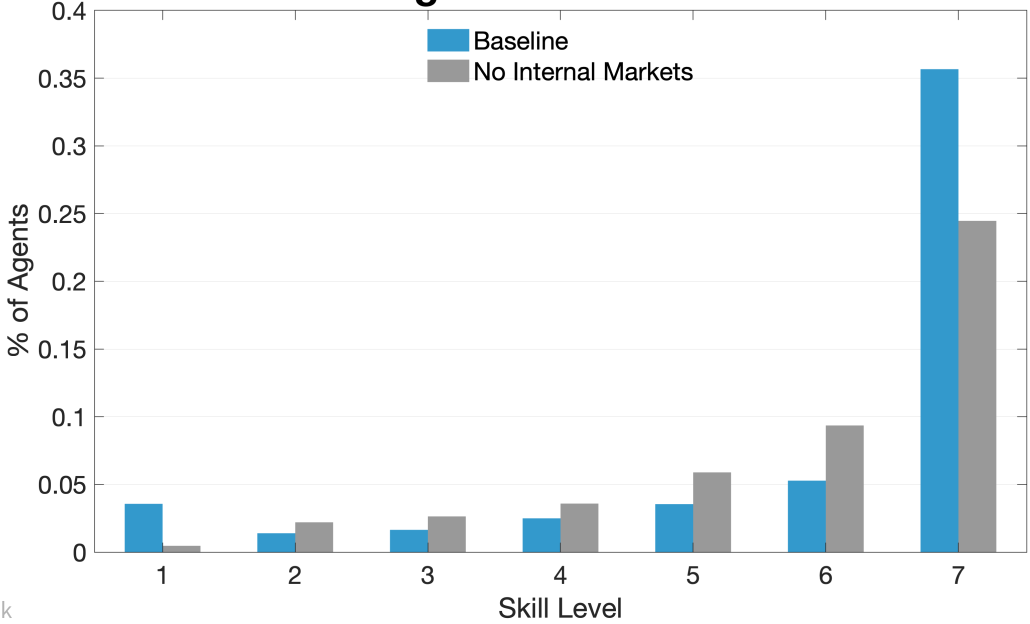
Manager Skill Distribution



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